

Simon Chen

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EDUCATION

Boston University

Boston, MA

Bachelor of Arts in Computer Science

Graduated: Spring 2026

- Relevant Coursework: Data Structures & Algorithms, Operating Systems, Distributed Systems, Databases, Machine Learning, Data Science.

EXPERIENCE

Software Engineering Intern

February 2026 – June 2026

RESET Standard

Shanghai, China

- Engineered asynchronous data pipelines with RabbitMQ, Redis, and Sidekiq to process long-running ingestion jobs in the background, decoupling external API calls from user-facing requests.
- Doubled the platform's supported environmental data categories (from 2 to 4) by independently implementing end-to-end water and waste data integration, developing Rails backend ingestion pipelines, PostgreSQL schemas, and client-facing React/Vite/Tailwind dashboards.
- Redesigned project-device mapping workflows in response to client requests, refactoring legacy forms to streamline onboarding of environmental monitoring devices.

Lead Product Engineer

December 2025 – Present

EZ Esports

New York, NY

- Designed a PostgreSQL schema on Supabase using Drizzle ORM, supporting 500+ matches across 20+ teams, and built RESTful endpoints for player statistics, standings, and historical data.
- Cut page load time by 75% (from 3.2s to 0.8s) by implementing edge caching on Cloudflare, route-based code splitting, and asset optimization for the Next.js frontend.
- Migrated an esports league platform from Google Sites to Next.js with a Supabase backend, replacing manual content updates with live scoring and automated season management.

PROJECTS

Persephone | *Python, Cython/C++17, Rust, Modal, FAISS*

May 2026

- Rewrote 9 live hot-path kernels (order-book maintenance, order-book imbalance, trade intensity, event aggregation) as native Cython extensions, cutting Kalshi book maintenance from 21.8ms to 195μs p50 (112x).
- Designed a lock-free SPSC event ring in Cython/C++17 with cache-line-aligned atomic cursors and packed uint64 records, benchmarked at 584ns per enqueue and 703ns per event drained.
- Sped up 155K-row neighbor search from 758μs to 154μs p50 by caching FAISS IVF indexes and pinning FAISS threads to avoid OpenMP oversubscription.
- Cut backtest and parameter-sweep iteration from ~14 hours on laptop CPUs to ~15 minutes by moving tick backtests and sweeps to Modal NVIDIA GPU workers with FAISS GPU search and CuPy-vectorized math.
- Replaced KDE inference with a GBM theo layer, cutting model query latency from ~1.5ms to ~20μs per query (~75x) behind an unchanged live serving interface.
- Eliminated duplicate-send and crash-replay risk with a durable-before-wire order layer: intent journaled before dispatch, deterministic client IDs for idempotent retries, and persisted ack state for reconciliation.

Self-Hosted Server Infrastructure | *Docker, Linux, Cloudflare, Nginx, Automation*

November 2025

- Architected containerized home-server infrastructure using Docker Compose, orchestrating 10+ self-hosted services for media streaming, automated library management, and monitoring
- Configured automated workflow pipelines with inter-container networking and Nginx reverse proxy routing, establishing centralized monitoring for resource optimization
- Implemented secure external access using Cloudflare Tunnels with zero-trust network architecture, eliminating port forwarding vulnerabilities while hosting personal website and services accessible via custom domain

TECHNICAL SKILLS

Languages: Python, C++17, Cython, Rust, SQL, TypeScript, JavaScript, Java, Ruby

Frameworks & Libraries: FAISS, CuPy, FastAPI, Django, Spring Boot, React, Next.js, Node.js, Vite, React Native, Ruby on Rails, Tailwind CSS

Databases & Infrastructure: Docker, OpenMP, PostgreSQL, Redis, RabbitMQ, Modal, Nginx, Cloudflare, Sidekiq, SQLite, Supabase